

Day 20 Research Document

Project Title

AI-Based Smart Agricultural Drone for Precision Farming

Introduction

- Modern agriculture increasingly relies on advanced technologies to improve productivity while reducing resource consumption.
- Drones equipped with AI and sensors enable farmers to monitor crops quickly and accurately across large agricultural fields.
- Real-time aerial monitoring helps identify crop stress, pest infestations, and irrigation issues before they become severe.
- This proposed solution combines drone technology, artificial intelligence, and precision farming to improve crop management.

Problem Statement

- Farmers often detect crop diseases only after visible damage has spread, leading to reduced agricultural productivity.
- Traditional field inspections are labor-intensive, time-consuming, and difficult to perform frequently on large farms.
- Uniform spraying of pesticides and fertilizers results in unnecessary chemical usage, increased costs, and environmental pollution.
- Limited access to real-time crop health information makes it difficult for farmers to take timely corrective actions.

Proposed Solution

- An autonomous drone equipped with AI cameras and environmental sensors performs regular aerial surveys of farmland.
- The onboard AI analyzes captured images to identify unhealthy crops, pest attacks, and nutrient deficiencies.
- The drone performs precision spraying only in affected areas, reducing chemical waste and improving farming efficiency.
- Collected data is uploaded to a cloud dashboard where farmers can monitor crop conditions and receive recommendations.

Objectives

- Develop an intelligent drone system capable of monitoring crop health with minimal human intervention.
- Reduce pesticide and fertilizer consumption through targeted precision spraying techniques.
- Provide farmers with real-time crop health reports for faster and better decision-making.
- Improve agricultural productivity while reducing operational costs and environmental impact.

Expected Outcomes

- Early detection of crop diseases helps prevent large-scale crop damage and increases overall yield.
- Precision spraying minimizes unnecessary chemical application, reducing costs and protecting the environment.
- Real-time monitoring improves farm management by providing accurate and timely agricultural insights.
- The proposed system supports sustainable farming practices through efficient use of resources and smart automation.

Conclusion

- The proposed AI-based agricultural drone offers a practical solution for modern precision farming challenges.
- Combining AI, IoT, and drone technologies improves monitoring accuracy and operational efficiency.
- The system has the potential to increase crop productivity while reducing labor and resource consumption.
- This concept demonstrates how innovative drone technology can support sustainable and intelligent agriculture.